

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Complete set of values of K in order that $f(x) = \sin x - \cos x - Kx + b$ decreases for all real values is given by -
 (A) $K < 1$ (B) $K \geq 1$ (C) $K \geq \sqrt{2}$ (D) $K < \sqrt{2}$ MMN001
2. If the function $f(x) = 2x^2 - kx + 5$ is strictly increasing in $[1, 2]$, then complete set of values of k is
 (A) $(-\infty, 4)$ (B) $[4, \infty)$ (C) $(-\infty, 8]$ (D) $[8, \infty)$ MMN002
3. If $f(x) = x^3 - 10x^2 + 200x - 10$, then $f(x)$ is-
 (A) strictly decreasing in $(-\infty, 10]$ and strictly increasing in $[10, \infty)$
 (B) strictly increasing in $(-\infty, 10]$ and strictly decreasing in $[10, \infty)$
 (C) strictly increasing for every value of x
 (D) strictly decreasing for every value of x MMN004
4. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in -
 (A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$ (C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$ MMN012
5. A function is matched below against an interval where it is supposed to be increasing. which of the following pairs is incorrectly matched ?
- | interval | function |
|---|-------------------------|
| (A) $(-\infty, \infty)$ | $x^3 - 3x^2 + 3x + 3$ |
| (B) $[2, \infty)$ | $2x^3 - 3x^2 - 12x + 6$ |
| (C) $\left(-\infty, \frac{1}{3}\right]$ | $3x^2 - 2x + 1$ |
| (D) $(-\infty, -4)$ | $x^3 + 6x^2 + 6$ |
- MMN015
6. If function $f(x) = 2x^2 + 3x - m \ln x$ is monotonic decreasing in the interval $(0, 1)$, then the least value of the parameter m is -
 (A) 7 (B) $\frac{15}{2}$ (C) $\frac{31}{4}$ (D) 8 MMN003
7. If $f: [1, 10] \rightarrow [1, 10]$ is a non-decreasing function and $g: [1, 10] \rightarrow [1, 10]$ is a non-increasing function. Let $h(x) = f(g(x))$ with $h(1) = 1$, then $h(2)$
 (A) lies in $(1, 2)$ (B) is more than 2 (C) is equal to 1 (D) is not defined MMN014
8. If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval-
 (A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) none MMN005

9. If the function $f(x) = x^3 - 6x^2 + ax + b$ defined on $[1, 3]$, satisfies the Rolle's theorem for $c = \frac{2\sqrt{3}+1}{\sqrt{3}}$, then -

(A) $a = 11, b = 6$ (B) $a = -11, b = 6$ (C) $a = 11, b \in R$ (D) None of these

MMN007

10. The equation $\sin x + x \cos x = 0$ has at least one root in

(A) $\left(-\frac{\pi}{2}, 0\right)$ (B) $(0, \pi)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

MMN011

11. A value of c for which the conclusion of Mean values theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is -

(A) $2\log_3 e$ (B) $\frac{1}{2} \log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$

MMN006

12. Given: $f(x) = 4 - \left(\frac{1}{2} - x\right)^{2/3}$; $g(x) = \begin{cases} \frac{\tan[x]}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

$$h(x) = \{x\}; k(x) = 5^{\log_2(x+3)}$$

then in $[0, 1]$, Lagrange's Mean Value Theorem is NOT applicable to

(A) f, g, h (B) h, k (C) f, g (D) g, h, k

where $[x]$ and $\{x\}$ denotes the greatest integer and fractional part function.

MMN008

13. Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x) \forall x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals

(A) $-1/4$ (B) $-1/8$ (C) $1/4$ (D) $1/8$

MMN009

14. If the function $f(x) = 2x^2 + 3x + 5$ satisfies LMVT at $x = 2$ on the closed interval $[1, a]$, then the value of 'a' is equal to

(A) 3 (B) 4 (C) 6 (D) 1

MMN010

15. The function $f(x) = (x - 3)^2$ satisfies all the conditions of Lagrange's mean value theorem in $[3, 4]$. A point on $y = (x - 3)^2$, where the tangent is parallel to the chord joining $(3, 0)$ and $(4, 1)$ is

(A) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (C) $(1, 4)$ (D) $(4, 1)$

MMN013

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Which of the following statements is/are correct?
 (A) $x + \sin x$ is increasing function
 (B) $\sec x$ is neither increasing nor decreasing function
 (C) $x + \sin x$ is decreasing function
 (D) $\sec x$ is an increasing function
MMN037
2. If $f(x) = 2x + \cot^{-1}x + \ln(\sqrt{1+x^2} - x)$, then $f(x)$:
 (A) increases in $[0, \infty)$ (B) decreases in $[0, \infty)$
 (C) neither increases nor decreases in $[0, \infty)$ (D) increases in $(-\infty, \infty)$
MMN038
3. Let $f(x) = \sin^2 x$ & $0 < A < B$, where $A, B \in \left(0, \frac{\pi}{4}\right)$, then
 (A) $(B - A) \sin 2A < \sin^2 B - \sin^2 A < (B - A) \sin 2B$
 (B) $(B - A) \sin 2A > \sin^2 B - \sin^2 A > (B - A) \sin 2B$
 (C) $\cos 2B < \cos 2A$
 (D) $B \cos 2A > A \cos 2B$
MMN042
4. Let $f(x)$ be a differentiable function and $f(\alpha) = f(\beta) = 0$ ($\alpha < \beta$), then in the interval (α, β)
 (A) $f(x) + f'(x) = 0$ has at least one root
 (B) $f(x) - f'(x) = 0$ has at least one real root
 (C) $f(x) \cdot f'(x) = 0$ has at least one real root
 (D) none of these
MMN041
5. Let $\phi(x) = (f(x))^3 - 3(f(x))^2 + 4f(x) + 5x + 3\sin x + 4\cos x \forall x \in R$, where $f(x)$ is a differentiable function $\forall x \in R$, then
 (A) ϕ is increasing whenever f is increasing
 (B) ϕ is increasing whenever f is decreasing
 (C) ϕ is decreasing whenever f is decreasing
 (D) ϕ is decreasing if $f'(x) = -11$
MMN040
6. Let $g(x) = 2f(x/2) + f(1-x)$ and $f''(x) < 0$ in $0 \leq x \leq 1$ then $g(x)$ ____
 (A) decreases in $\left[0, \frac{2}{3}\right]$ (B) decreases in $\left[\frac{2}{3}, 1\right]$
 (C) increases in $\left[0, \frac{2}{3}\right]$ (D) increases in $\left[\frac{2}{3}, 1\right]$
MMN039

7. For the function $f(x) = x \cot^{-1} x, x \geq 0$

(A) there is atleast one $x \in (0, 1)$ for which $\cot^{-1} x = \frac{x}{1+x^2}$

(B) for atleast one x in the interval $(0, \infty)$, $f\left(x + \frac{2}{\pi}\right) - f(x) < 1$

(C) number of solution of the equation $f(x) = \sec x$ is 1

(D) $f'(x)$ is strictly decreasing in the interval $(0, \infty)$

MMN106

8. Which of the following inequalities are valid -

(A) $|\tan^{-1} x - \tan^{-1} y| \leq |x - y| \forall x, y \in R$ (B) $|\tan^{-1} x - \tan^{-1} y| \geq |x - y|$

(C) $|\sin x - \sin y| \leq |x - y|$

(D) $|\sin x - \sin y| \geq |x - y|$

MMN117

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 9 to 11

If $f(x) = |x - 2| + |x - 4| + |x - 6|$.

9. If the set of values of x for which $f(x)$ increases is $[a + 2, \infty)$, then 'a' is -

MMN046

10. If the set of values of x for which $f(x)$ decreases is $(-\infty, b - 1]$, then 'b' is -

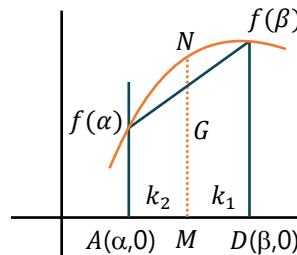
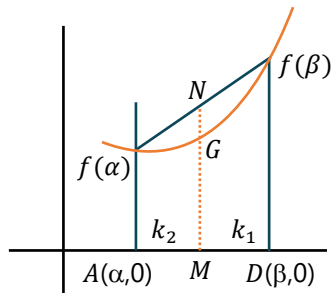
MMN047

11. If $f(x)$ is symmetrical about the line $x = K$, then K is equal to -

MMN048

Paragraph for Question 12 to 14

For a double differentiable function $f(x)$ if $f''(x) \geq 0$ then $f(x)$ is concave upward and if $f''(x) \leq 0$ then $f(x)$ is concave downward



Here $M\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}, 0\right)$

If $f(x)$ is a concave upward in $[a, b]$ and $\alpha, \beta \in [a, b]$ then $\frac{k_1 f(\alpha) + k_2 f(\beta)}{k_1 + k_2} \geq f\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}\right)$,

where $k_1, k_2 \in R^+$

If $f(x)$ is a concave downward in $[a, b]$ and $\alpha, \beta \in [a, b]$ then $\frac{k_1 f(\alpha) + k_2 f(\beta)}{k_1 + k_2} \leq f\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}\right)$,

where $k_1, k_2 \in R^+$

then answer the following:

12. Which of the following is true

- (A) $\frac{\sin \alpha + \sin \beta}{2} > \sin \left(\frac{\alpha + \beta}{2} \right); \alpha, \beta \in (0, \pi)$ (B) $\frac{\sin \alpha + \sin \beta}{2} < \sin \left(\frac{\alpha + \beta}{2} \right); \alpha, \beta \in (\pi, 2\pi)$
 (C) $\frac{\sin \alpha + \sin \beta}{2} < \sin \left(\frac{\alpha + \beta}{2} \right); \alpha, \beta \in (0, \pi)$ (D) None of these

MMN043

13. Which of the following is true

- (A) $\frac{2^\alpha + 2^{\beta+1}}{3} \leq 2^{\frac{\alpha+2\beta}{3}}$
 (B) $\frac{2\ell n \alpha + \ell n \beta}{3} \geq \ell n \left(\frac{2\alpha + \beta}{3} \right)$
 (C) $\frac{\tan^{-1} \alpha + \tan^{-1} \beta}{2} \leq \tan^{-1} \left(\frac{\alpha + \beta}{2} \right); a, b \in R^-$
 (D) $\frac{e^\alpha + 2e^\beta}{3} \geq e^{\frac{\alpha+2\beta}{3}}$

MMN044

14. Let α, β and γ are three distinct real numbers and $f''(x) < 0$. Also $f(x)$ is increasing function and let

$$A = \frac{f^{-1}(\alpha) + f^{-1}(\beta) + f^{-1}(\gamma)}{3} \text{ and } B = f^{-1} \left(\frac{\alpha + \beta + \gamma}{3} \right), \text{ then order relation between } A \text{ and } B \text{ is ?}$$

- (A) $A > B$ (B) $A < B$ (C) $A = B$ (D) none of these

MMN045

MATCHING LIST TYPE QUESTION

15. Match List-I with List-II and select the correct answer using the code given below the list.

List - I

(P) $f(x) = 2 \cdot e^{x^2-4x}$

(Q) $f(x) = \frac{e^x}{x}$

(R) $f(x) = x^2 e^{-x}$

(S) $f(x) = 2x^2 - \ln|x|$

List - II

(1) Strictly increasing in $(2, \infty)$ &
Strictly decreasing in $(-\infty, 2)$

(2) Strictly increasing in $(0, 2)$ &
Strictly decreasing in $(-\infty, 0) \cup (2, \infty)$

(3) Strictly increasing in $(1, \infty)$ &
Strictly decreasing in $(-\infty, 0) \cup (0, 1)$

(4) Strictly increasing in $\left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$ &
Strictly decreasing in $\left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

MMN049

EXERCISE (S)

1. If $f(x) = 2e^x - ae^{-x} + (2a + 1)x - 3$ increases for every $x \in \mathbb{R}$ then the range of values of 'a' is $[k, \infty)$ find k .

MMN016

2. Find the sum of value of a, m and b if the function $f(x) = \begin{cases} 3 & x=0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \leq x \leq 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval $[0, 2]$.

MMN017

3. Find the least integral value of 'a' for which the function $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ increases throughout the number line.

MMN018

4. Let $f(x)$ be a strictly increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. The range of a is $\left(0, \frac{1}{p}\right) \cup (q, r)$. Find $(p + q + r)$.

MMN019

5. If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then $a \cdot f'(c) = b \cdot g'(c)$ for some $c \in (0, 1)$, then the value of $\frac{b}{a}$ is :

MMN020

6. If the sum of greatest & the least values of the function $f(x) = 12x^{\frac{4}{3}} - 6x^{\frac{1}{3}}$ in the interval $x \in [-1, 1]$, if they exist, is S then the value of $4S$ is

MMN050

7. Find the number of integral values of 'a' for which the function, $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ is strictly increasing at every point of its domain.

MMN051

8. Find the minimum value of the function $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x .

MMN052

9. If $b > a$ and the minimum value of $|x - a|^3 + |x - b|^3$, $x \in \mathbb{R}$, is $\frac{(b-a)^p}{k}$, then the value of $(k - p)$ is

MMN053

10. If the set of values of x for which the inequality $\ln(1+x) > x/(1+x)$ is valid is $(a, b) \cup (b, \infty)$, then the value of $(2a - b)$ is

MMN054

EXERCISE - JEE (Main) PYQ

1. If the function f given by $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7$, for some $a \in \mathbb{R}$ is increasing in $(0, 1]$ and decreasing in $[1, 5)$, then a root of the equation, $\frac{f(x)-14}{(x-1)^2} = 0 (x \neq 1)$ is : **[JEE (Main) 2019]**

(1) 6 (2) 5 (3) 7 (4) -7

MMN021

2. Let $f(x) = e^x - x$ and $g(x) = x^2 - x$, $\forall x \in \mathbb{R}$. Then the set of all $x \in \mathbb{R}$, where the function $h(x) = (f \circ g)(x)$ is increasing, is : **[JEE (Main) 2019]**

(1) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (2) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$ (3) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$ (4) $[0, \infty)$

MMN022

3. If m is the minimum value of k for which the function $f(x) = x\sqrt{kx - x^2}$ is increasing in the interval $[0, 3]$ and M is the maximum value of f in $[0, 3]$ when $k = m$, then the ordered pair (m, M) is equal to : **[JEE (Main) 2019]**

(1) $(4, 3\sqrt{2})$ (2) $(4, 3\sqrt{3})$ (3) $(3, 3\sqrt{3})$ (4) $(5, 3\sqrt{6})$

MMN023

4. The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is : **[JEE (Main) 2020]**

(1) $\frac{2}{3}$ (2) $\frac{\sqrt{7}-2}{3}$ (3) $\frac{4-\sqrt{5}}{3}$ (4) $\frac{4-\sqrt{7}}{3}$

MMN024

5. Let the function, $f : [-7, 0] \rightarrow \mathbb{R}$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$, for all $x \in (-7, 0)$, then for all such functions f , $f(-1) + f(0)$ lies in the interval : **[JEE (Main) 2020]**

(1) $[-6, 20]$ (2) $(-\infty, 20]$ (3) $(-\infty, 11]$ (4) $[-3, 11]$

MMN025

6. Let $f(x) = x \cos^{-1}(-\sin|x|)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then which of the following is true ? **[JEE (Main) 2020]**

(1) f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$

(2) f is not differentiable at $x = 0$

(3) $f'(0) = -\frac{\pi}{2}$

(4) f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

MMN026

7. Let f be any function continuous on $[a, b]$ and twice differentiable on (a, b) . If for all $x \in (a, b)$, $f'(x) > 0$ and $f''(x) < 0$, then for any $c \in (a, b)$, $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than:

[JEE (Main) 2020]

- (1) $\frac{b+a}{b-a}$ (2) $\frac{b-c}{c-a}$ (3) $\frac{c-a}{b-c}$ (4) 1

MMN027

8. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined as,

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4, \end{cases}$$

Let $A = \{x \in \mathbf{R} : f \text{ is increasing}\}$. Then A is equal to :

[JEE (Main) 2021]

- (1) $(-\infty, -5) \cup (4, \infty)$ (2) $(-5, \infty)$
(3) $(-\infty, -5) \cup (-4, \infty)$ (4) $(-5, -4) \cup (4, \infty)$

MMN028

9. If Rolle's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$, $x \in [1, 2]$ with $f'\left(\frac{4}{3}\right) = 0$, then ordered pair (a, b) is equal to :

[JEE (Main) 2021]

- (1) (5, 8) (2) (-5, 8) (3) (5, -8) (4) (-5, -8)

MMN029

10. Consider the function $f : \mathbf{R} \rightarrow \mathbf{R}$ defined by $f(x) = \begin{cases} \left(2 - \sin\left(\frac{1}{x}\right)\right)|x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then f is :

- (1) monotonic on $(-\infty, 0) \cup (0, \infty)$ (2) not monotonic on $(-\infty, 0)$ and $(0, \infty)$
(3) monotonic on $(0, \infty)$ only (4) monotonic on $(-\infty, 0)$ only

[JEE (Main) 2021]

MMN030

11. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ and $g : \mathbf{R} \rightarrow \mathbf{R}$ be two functions defined by $f(x) = \log_e(x^2 + 1) - e^{-x} + 1$ and $g(x) = \frac{1 - 2e^{2x}}{e^x}$. Then, for which of the following range of α , the inequality

$$f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right) \text{ holds?}$$

[JEE (Main) 2022]

- (1) (2, 3) (2) (-2, -1) (3) (1, 2) (4) (-1, 1)

MMN031

12. The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____. [JEE (Main) 2022]

- (1) 0 (2) 1 (3) 3 (4) 5

MMN032

13. The function $f(x) = xe^{x(1-x)}, x \in R$, is [JEE (Main) 2022]

- (1) increasing in $\left(-\frac{1}{2}, 1\right)$ (2) decreasing in $\left(\frac{1}{2}, 2\right)$
 (3) increasing in $\left(-1, -\frac{1}{2}\right)$ (4) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

MMN033

14. Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \log_e(\sqrt{1+x^2} + x), x \in [0, 3]$. Then [JEE (Main) 2023]

- (1) There exists $x \in [0, 3]$ such that $f'(x) < g'(x)$
 (2) $\max f(x) > \max g(x)$
 (3) There exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x), \forall x \in (x_1, x_2)$
 (4) $\min f'(x) = 1 + \max g'(x)$

MMN034

15. Let $f: (0, 1) \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \frac{1}{1-e^{-x}}, \text{ and } g(x) = (f(-x) - f(x)). \text{ Consider two statements}$$

- (I) g is an increasing function in $(0, 1)$
 (II) g is one-one in $(0, 1)$

Then,

- (1) Only (I) is true (2) Only (II) is true
 (3) Neither (I) nor (II) is true (4) Both (I) and (II) are true

[JEE (Main) 2023]

MMN035

16. Let $g(x) = f(x) + f(1-x)$ and $f''(x) > 0, x \in (0, 1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{-1}(2\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right) + \tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to :

[JEE (Main) 2023]

- (1) $\frac{3\pi}{2}$ (2) π (3) $\frac{5\pi}{4}$ (4) $\frac{3\pi}{4}$

MMN036

17. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeros of $(3f'f'' + ff''')(x)$ is

[JEE (Main) 2024]

MMN118

18. For the function $f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x)$, where $x \in \left[0, \frac{\pi}{2}\right]$, consider the following two statements :

- (I) f is increasing in $\left(0, \frac{\pi}{2}\right)$.
 (II) f' is decreasing in $\left(0, \frac{\pi}{2}\right)$.

Between the above two statements,

- (1) only (I) is true. (2) only (II) is true.
 (3) neither (I) nor (II) is true. (4) both (I) and (II) are true.

[JEE (Main) 2024]

MMN119

19. Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for all $x \in (0, 3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is
- (1) 24 (2) 0 (3) 18 (4) 20

[JEE (Main) 2024]

MMN120

20. Consider the function $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ defined by $f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$. Consider the statements
- (I) The curve $y = f(x)$ intersects the x -axis exactly at one point
- (II) The curve $y = f(x)$ intersects the x -axis at $x = \cos \frac{\pi}{12}$

Then

[JEE (Main) 2024]

- (1) Only (II) is correct (2) Both (I) and (II) are incorrect
- (3) Only (I) is correct (4) Both (I) and (II) are correct

MMN121

21. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a non constant twice differentiable such that $g'\left(\frac{1}{2}\right) = g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x) = \frac{1}{2}[g(x) + g(2-x)]$, then

[JEE (Main) 2024]

- (1) $f''(x) = 0$ for atleast two x in $(0, 2)$
- (2) $f''(x) = 0$ for exactly one x in $(0, 1)$
- (3) $f''(x) = 0$ for no x in $(0, 1)$
- (4) $f'\left(\frac{3}{2}\right) + f'\left(\frac{1}{2}\right) = 1$

MMN122

EXERCISE - JEE (Advanced) PYQ

1. Let $f : (0,1) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{b-x}{1-bx}$, where b is a constant such that $0 < b < 1$. Then
[JEE (Advanced) 2011]
(A) f is not invertible on $(0,1)$ (B) $f \neq f^{-1}$ on $(0,1)$ and $f'(b) = \frac{1}{f'(0)}$
(C) $f = f^{-1}$ on $(0,1)$ and $f'(b) = \frac{1}{f'(0)}$ (D) f^{-1} is differentiable on $(0,1)$
MMN055
2. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is
[JEE (Advanced) 2011]
MMN056
3. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is
[JEE (Advanced) 2013]
(A) 6 (B) 4 (C) 2 (D) 0
MMN057
4. Let $f(x) = x \sin \pi x$, $x > 0$. Then for all natural numbers n , $f'(x)$ vanishes at -
[JEE (Advanced) 2013]
(A) a unique point in the interval $\left(n, n + \frac{1}{2}\right)$ (B) a unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$
(C) a unique point in the interval $(n, n + 1)$ (D) two points in the interval $(n, n + 1)$
MMN058
5. Let $a \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^5 - 5x + a$. Then
[JEE (Advanced) 2014]
(A) $f(x)$ has three real roots if $a > 4$ (B) $f(x)$ has only one real roots if $a > 4$
(C) $f(x)$ has three real roots if $a < -4$ (D) $f(x)$ has three real roots if $-4 < a < 4$
MMN059
6. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by $f(x) = (\log(\sec x + \tan x))^3$. Then :-
[JEE (Advanced) 2014]
(A) $f(x)$ is an odd function (B) $f(x)$ is a one-one function
(C) $f(x)$ is an onto function (D) $f(x)$ is an even function
MMN060
7. Let $f, g : [-1, 2] \rightarrow \mathbb{R}$ be continuous function which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table :
- | | $x = -1$ | $x = 0$ | $x = 2$ |
|--------|----------|---------|---------|
| $f(x)$ | 3 | 6 | 0 |
| $g(x)$ | 0 | 1 | -1 |
- In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is(are)
[JEE (Advanced) 2015]
(A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$
(B) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(-1, 0)$
(C) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(0, 2)$
(D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$
MMN061

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table. [JEE (Advanced) 2017]

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$.

- * Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.
- * Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.
- * Column 3 contains information about increasing/decreasing nature of $f(x)$ and $f'(x)$.

Column 1	Column 2	Column 3
(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$
(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III) $f'(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$	(R) f' is increasing in $(0, 1)$
(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x \rightarrow \infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

8. Which of the following options is the only **CORRECT** combination ?

- (A) (IV) (i) (S) (B) (I) (ii) (R) (C) (III) (iv) (P) (D) (II) (iii) (S)

MMN062

9. Which of the following options is the only **CORRECT** combination ?

- (A) (III) (iii) (R) (B) (I) (i) (P) (C) (IV) (iv) (S) (D) (II) (ii) (Q)

MMN063

10. Which of the following options is the only **INCORRECT** combination ?

- (A) (II) (iii) (P) (B) (II) (iv) (Q) (C) (I) (iii) (P) (D) (III) (i) (R)

MMN064

11. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and

$$f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1, \text{ then}$$

[JEE (Advanced) 2017]

- (A) $0 < f'(1) \leq \frac{1}{2}$ (B) $f'(1) \leq 0$ (C) $f'(1) > 1$ (D) $\frac{1}{2} < f'(1) \leq 1$

MMN065

12. For a polynomial $g(x)$ with real coefficient, let m_g denote the number of distinct real roots of $g(x)$. Suppose S is the set of polynomials with real coefficients defined by

$$S = \{(x^2 - 1)^2 (a_0 + a_1 x + a_2 x^2 + a_3 x^3) : a_0, a_1, a_2, a_3 \in \mathbb{R}\}.$$

For a polynomial f , let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of $(m_{f'} + m_{f''})$, where $f \in S$, is ____ [JEE (Advanced) 2020]

MMN066

13. $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is **TRUE** ? [JEE (Advanced) 2020]

- (A) f is one-one, but **NOT** onto
 (B) f is onto, but **NOT** one-one
 (C) f is **BOTH** one-one and onto
 (D) f is **NEITHER** one-one **NOR** onto

MMN067

14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}.$$

Then which of the following statements is (are) TRUE ?

[JEE (Advanced) 2021]

- (A) f is decreasing in the interval $(-2, -1)$ (B) f is increasing in the interval $(1, 2)$
 (C) f is onto (D) Range of f is $\left[-\frac{3}{2}, 2\right]$

MMN068

15. Let S be the set of all twice differentiable functions f from $\mathbb{R}$ to $\mathbb{R}$ such that $\frac{d^2 f}{dx^2}(x) > 0$ for all $x \in (-1, 1)$. For $f \in S$, let X_f be the number of points $x \in (-1, 1)$ for which $f(x) = x$. Then which of the following statements is (are) true?

[JEE (Advanced) 2023]

- (A) There exists a function $f \in S$ such that $X_f = 0$
 (B) For every function $f \in S$, we have $X_f \leq 2$
 (C) There exists a function $f \in S$ such that $X_f = 2$
 (D) There does **NOT** exist any function f in S such that $X_f = 1$

MMN123

16. Let the function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{\sin x (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)} + \frac{2 (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)}.$$

Then the number of solutions of $f(x) = 0$ in $\mathbb{R}$ is ____.

[JEE (Advanced) 2024]

MMN124

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	C	B	C	A	C	A	C	B
Que.	11	12	13	14	15					
Ans.	A	A	B	A	B					

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,D	A,C,D	A,B,C	A,D	B,C	B,D	A,C	2	5
Que.	11	12	13	14	15					
Ans.	4	C	D	A	B					

EXERCISE (S)

1.	0	2.	8	3.	1	4.	9	5.	3
6.	63	7.	8	8.	-10	9.	1	10.	-2

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	4	2	1	3	4	1	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	1	2	4	2	5	4	3	4
Que.	21									
Ans.	1									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	2	C	B,C	B,D	A,B,C	B,C	D	D	D
Que.	11	12	13	14	15	16				
Ans.	C	5	C	A,B	A,B,C	1				